Problem 1.

证明：为书写简便，记：$f_0 = f(x=0, u)$，$f_1 = f(x=1, u)$

$g_0 = g(x=0, u)$，$g_1 = g(x=1, u)$.

因此：$D_x f = f_0 \oplus f_1$，$D_x g = g_0 \oplus g_1$

由于原式左边为与 x 无关的函数，右边为与 x 有关的函数，考虑枚举 x 的取值分类证明

① 当 $x=0$ 时：

右边 $= f_0(g_0 \oplus g_1) \oplus g_0(f_0 \oplus f_1) \oplus (f_0 \oplus f_1)(g_0 \oplus g_1)$

由于：$a(b \oplus c) = a\bar{b}c + ab\bar{c} = ab \oplus ac$

因此：右边 $= f_0 g_0 \oplus f_0 g_1 \oplus g_0(f_0 \oplus f_1) \oplus g_0(f_0 \oplus f_1) \oplus g_1(f_0 \oplus f_1)$.

由于：$a \oplus a = 0$，$0 \oplus a = a$. 故：

右边 $= f_0 g_0 \oplus f_0 g_1 \oplus f_0 g_1 \oplus f_1 g_1 = f_0 g_0 \oplus f_1 g_1$

而左边 $= D_x(fg) = f_0 g_0 \oplus f_1 g_1$，

因此 $x=0$ 时，原式成立，

③当 $x=1$ 时：左边化为 $fg_0 \oplus fg_1$

右边 $= f_1(g_0 \oplus g_1) \oplus g_1(f_0 \oplus f_1) \oplus g_0(f_0 \oplus f_1) \oplus g_1(f_0 \oplus f_1)$

$= fg_0 \oplus fg_1 \oplus g_0 f_0 \oplus g_0 f_1 = fg_0 \oplus fg_1 =$ 左边

综上，原式成立，证毕.

Problem 2

(a). 证明：布尔函数有 $\wedge$ (AND), $\vee$ (OR), $\neg$ (NOT) 三种.

AND: $a \wedge b = \neg(\neg(a \wedge b)) = \neg a \uparrow b$.

OR: $a \vee b = \neg(\neg(a \vee b)) = \neg[(\neg a) \wedge (\neg b)]$

$= (\neg a) \uparrow (\neg b)$

NOT: $\neg a = \neg(a \wedge a) = a \uparrow a$.

因此 NAND 是完备的，证毕.

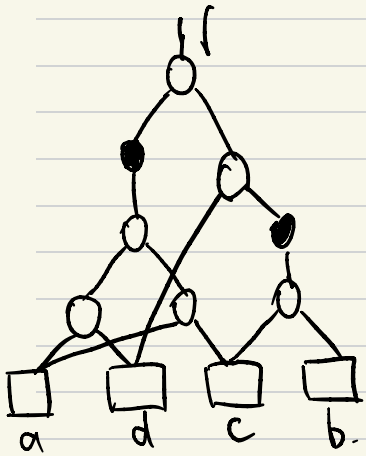
(b) $f(a,b,c,d) = (a \wedge d) \vee (b \wedge c \wedge d) \vee (a \wedge c)$

$= \neg(a \uparrow d \wedge \neg(b \wedge c \wedge d) \wedge a \uparrow c)$

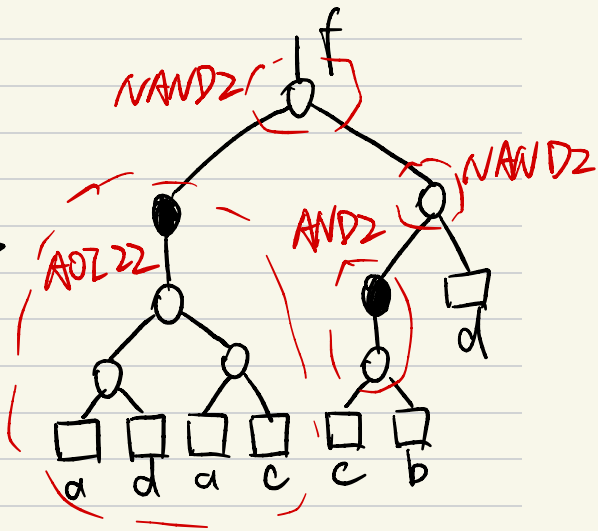
$$= \neg(a \uparrow d \wedge (b \wedge c) \uparrow d \wedge a \uparrow c)$$

$$= \{\neg[(a \uparrow d) \uparrow (a \uparrow c)]\} \uparrow [(\neg(b \uparrow c)) \uparrow d]$$

(c) f对应的AIG为：



Treesifying



红色标注即最佳 tree covering

共需 2个NAND2，1个AND2，

1个AOI22

$$ad + ac + bcd$$

$$= \overline{(\bar{a}+\bar{d})(\bar{a}+\bar{c})(\bar{b}+\bar{c}+\bar{d})}$$

$$= \overline{\bar{a}\bar{d} \cdot \bar{a}\bar{c} \cdot \overline{bcd} \cdot \bar{a}\bar{d}}$$

$$=$$